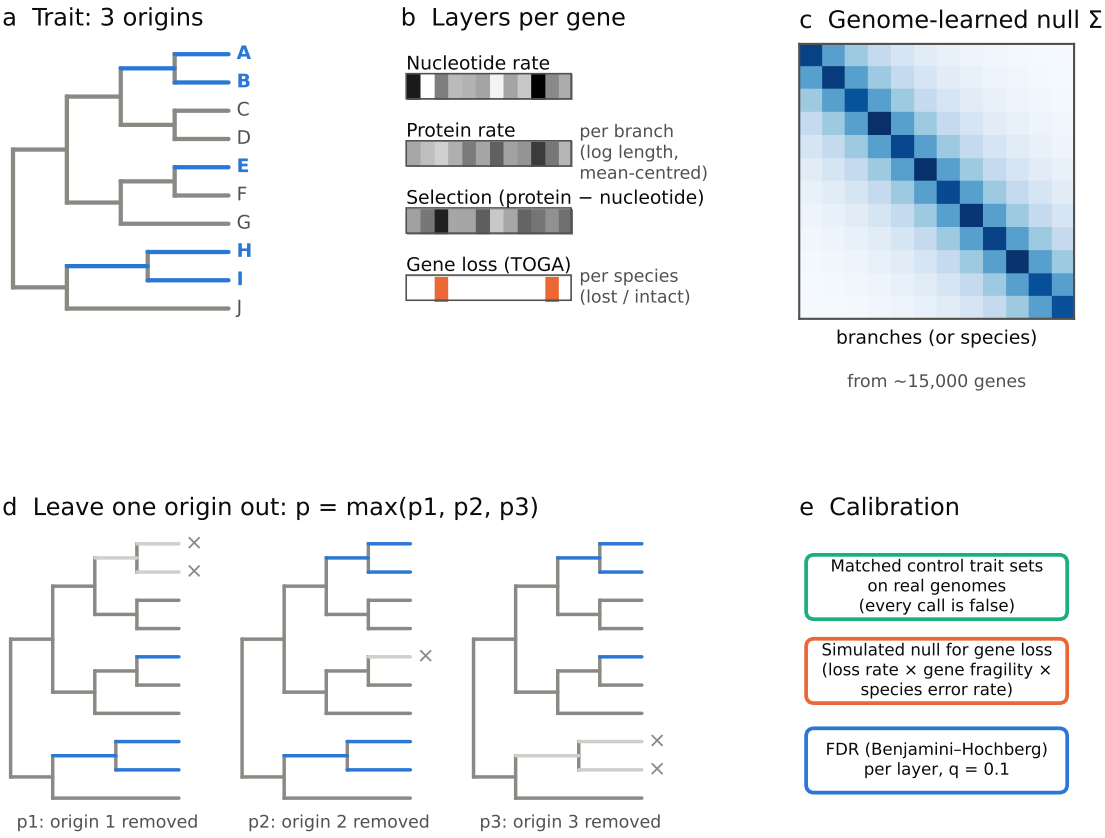
a Trait: 3 origins
A
B
C
D
E
F
G
H
I
J
b Layers per gene
Nucleotide rate
Protein rate
per branch
(log length,
mean-centred)
Selection (protein − nucleotide)
Gene loss (TOGA)
per species
(lost / intact)
c Genome-learned null Σ
branches (or species)
from ~15,000 genes
d Leave one origin out: p = max(p1, p2, p3)
p1: origin 1 removed
p2: origin 2 removed
p3: origin 3 removed
e Calibration
Matched control trait sets
on real genomes
(every call is false)
Simulated null for gene loss
(loss rate × gene fragility ×
species error rate)
FDR (Benjamini–Hochberg)
per layer, q = 0.1